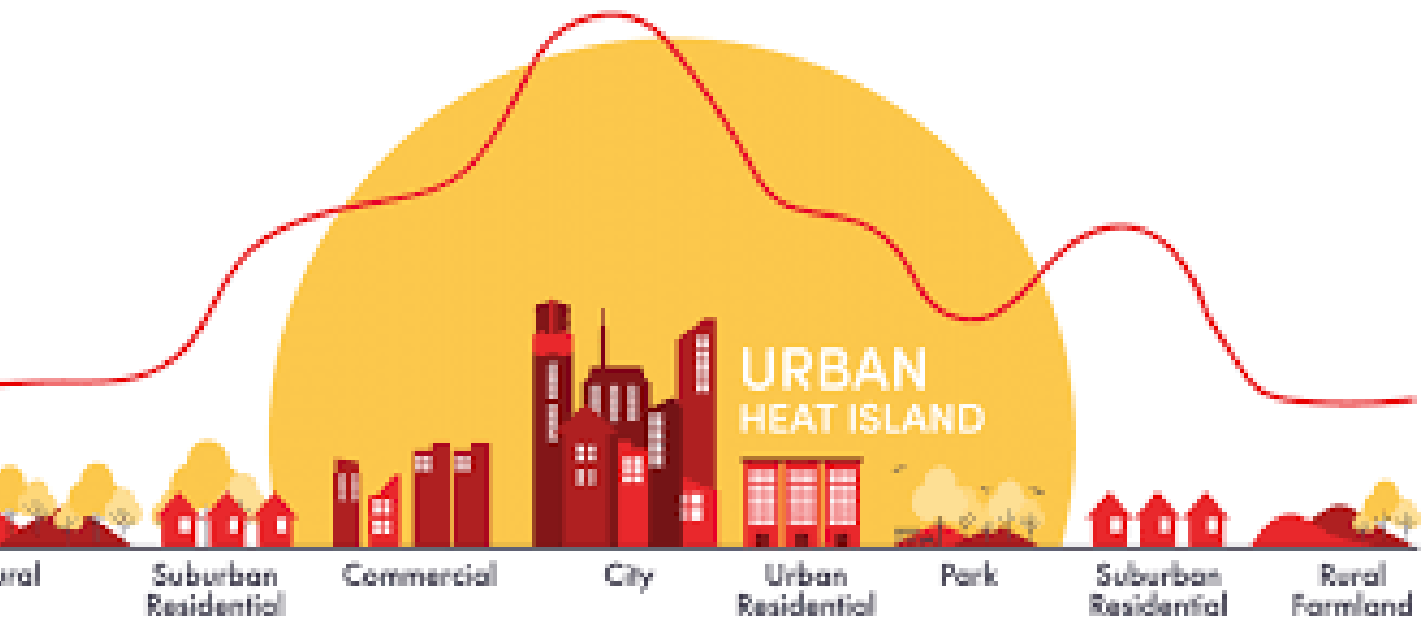




Urban heat island

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1. Why is the Urban Heat Island a problem?

The urban heat island is mainly a problem for local residents. It causes many health risks, such as air pollution, which is partly due to the formation of more ozone as a result of higher temperatures. It also creates heat stress, which affects the sleep quality of residents, especially those living in the city centre. These health risks are particularly serious for the elderly, young children, and people with vulnerable health.

In addition, the urban heat island affects the liveability of the city in terms of living, working, and recreation.

Energy costs are higher for people living in the city compared to those in surrounding areas. In summer, asphalt and concrete retain a lot of heat. As a result, temperatures remain significantly higher at night. Because it stays warmer at night, people use air conditioning more, which leads to higher energy consumption.

2. How did the problem arise in Alphen aan den Rijn?

It has been a problem in Alphen aan den Rijn for a long time. The problem arose after the city was built and many people moved there. During the construction of the city, too little consideration was given to creating green spaces. The houses were also not built strategically. As they are built now, there is little airflow through the city, which would otherwise help to cool it down.

Alphen aan den Rijn has also made extensive use of “heavy” materials instead of lighter ones. Asphalt and concrete retain heat. This is especially disadvantageous during summer nights, because it stays warm and the city cools down less.

In short, we think the problem already started during the construction of Alphen aan den Rijn, because too little attention was paid to this issue at the time. Over the years, the problem has only become worse.

3. What is Alphen aan den Rijn doing about it?

In Alphen aan den Rijn, a first step has been taken towards a local heat plan, which must become a concrete and workable plan before April 2026. Residents are already being informed and asked to identify locations that could serve as cool places. Alphen aan den Rijn is constructing underground water storage systems, and a lot of greenery is being planted in the city centre.

Residents are encouraged to participate in “Steenbreek,” a campaign in which people replace tiles with greenery. This includes:

1. Involvement of social organisations, care institutions, schools, and emergency services
2. Support for vulnerable residents living at home
3. Opening air-conditioned public locations

For the municipality, it is especially important that residents understand what measures they can take themselves and that they know what the municipality (including organisations) is doing for them.

4. What more could Alphen aan den Rijn do?

We believe that Alphen aan den Rijn could create more green spaces in the city. We think there is especially potential on the roofs of houses and buildings. At the moment, there is hardly any greenery on rooftops, but if Alphen aan den Rijn were to invest in green roofs, this would help reduce the urban heat island effect.

It would also help to make residents even more aware of the problem and encourage them to participate more actively in the Steenbreek campaign. In many gardens in Alphen aan den Rijn, there is still too much stone that could be replaced with greenery.

Finally, it is important that Alphen aan den Rijn starts using different materials and replaces old ones. A lot of construction in Alphen currently uses concrete and asphalt. These “heavy” materials absorb and retain a lot of heat, while lighter and more reflective materials do not. Alphen could also use a lighter asphalt mixture that absorbs less heat.

We therefore believe that Alphen should pay more attention to the materials it uses.